

ERROR ANALYSIS FOR THE INTERPOLATION-SUPPLEMENTED LATTICE-BOLTZMANN EQUATION SCHEME

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The numerical error of the interpolation-supplemented lattice-Boltzmann equation (ISLBE) scheme is theoretically analyzed. The ISLBE is proved to recover the Navier–Stokes equation as long as a second order interpolation scheme is used and the flow domain is appropriately discretized. Using a linear interpolation scheme will inevitably generate the numerical diffusivity and viscosity, making the ISLBE scheme unable to simulate the Navier–Stokes equation correctly. The analysis is consistent with previous numerical experiments.

Keywords: Lattice-Boltzmann Equation; Non-uniform Grid; Numerical Error.

1. Introduction

As a promising new computational fluid dynamics (CFD) approach, the lattice-Boltzmann equation (LBE) method has attracted much attention in engineering and computational physics communities in recent years.¹ Unlike its precursor — the lattice-gas automata (LGA),^{2,3} the LBE method uses a single particle distribution function instead of the Boolean fluid model.⁴ This replacement eliminates the time-consuming statistical average step, and yet retains the virtue of parallel computing. Introduction of the single-time relaxation BGK model⁵ for the collisional operator further simplifies the algorithm, enhances the computational efficiency, and more importantly, eliminates the lack of Galilean invariance and dependence of pressure on velocity.^{6,7} Theoretically, the lattice-Boltzmann equation can be proved to recover the Navier–Stokes equation under the Chapman–Enskog expansion.⁸ Quantitative studies^{8,9} have shown that the LBE method is computationally more stable and efficient than conventional CFD methods with a competitive accuracy.

Despite its exciting success in simulations of complicated flows, the previous lattice-Boltzmann equation models were plagued with an undesirable feature — the restriction to regular grids in physical space.¹⁰ The computational grids in previous LBE models can only have a triangular or square shape in a two-dimensional domain

and a cubic shape in a three-dimensional domain. Undoubtedly, this restriction greatly hampers the wide applications of the LBE method to practical problems. This drawback is not a surprise considering that the LBE method evolves directly from the LGA, in which the velocity space must be discretized into a regular lattice to guarantee isotropy. Due to the Boolean nature of the lattice-gas automata, the collision in LGA must take place at grid sites. Consequently, particles have to hop from one grid site to another in one time step. In other words, in LGA, the discretization of physical space is coupled with the discretization of velocity space. Although the previous LBE models differ from the LGA model in many respects, the restriction to regular lattices remained.

The coupling between the discretization of velocity and physical spaces is essential for the LGA models, but not necessary for the LBE method. This argument originates from a recent discovery^{11,12} that, besides the LGA, the LBE method can be derived from kinetic theory and the associated Boltzmann equation. Under this framework, it is clear that the lattice-Boltzmann equation is nothing but a special discrete form of the Boltzmann equation. Based on the above facts, the author and colleagues^{13–15} have proposed an interpolation-supplemented lattice-Boltzmann equation (ISLBE) scheme which can be implemented on an irregular grid. In this new scheme, the discretization of the physical space is freed from the discretization of velocity space, and therefore the computational mesh can be arbitrary. The only modification in the ISLBE scheme is the introduction of a new interpolation step, which is necessary for reconstructing the information on each grid site after each streaming process.

Numerical simulations using the ISLBE scheme have been carried out on flow in a sudden expansion channel^{13,14} and flow around a cylinder.¹⁵ These studies adopted typical variable grids used in other traditional CFD simulations, and still were able to obtain results with good agreement with existing experimental measurements and other CFD simulation results. Of course, the accuracy of the ISLBE scheme appears to depend strongly on the interpolation scheme and the computational mesh.¹³ To ensure the numerical error is in an acceptable range, a second order interpolation scheme is necessary and flow domains have to be discretized appropriately.

To illustrate the accuracy of this new computational scheme and its applicable range, a numerical test itself is not sufficient and a theoretical analysis is always desirable. It is the purpose of this paper to carry out such an analysis of the numerical errors in the ISLBE scheme. This analysis will serve as a theoretical justification for the ISLBE scheme. The rest of this paper is organized as follows. Section 2 briefly describes the ISLBE scheme for the sake of completeness. Section 3 analyzes the numerical error of the ISLBE scheme for both linear and quadratic interpolations. Section 4 concludes the paper.

2. Interpolation-Supplemented Lattice-Boltzmann Equation Scheme

In principle, the ISLBE scheme is applicable to problems in any dimension. For simplicity, however, this study will only use the one-dimensional model as an

example to illustrate the numerical error in the ISLBE scheme. Figure 1 shows a typical mesh setup in the ISLBE scheme. A flow domain is discretized into a non-uniform grid mesh with grid nodes of $(x_1, x_2, \dots, x_n)$. The grid size is denoted by $\Delta_{i+\frac{1}{2}} = x_{i+1} - x_i$ which is not necessarily a constant. At each grid node sits a regular velocity lattice, i.e., a 3-bit lattice (e_0, e_1, e_2) .

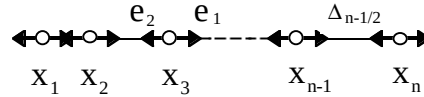


Fig. 1. Schematic plot of the lattice-Boltzmann method on a variable grid.

The ISLBE scheme consists of three steps: collision, streaming, and interpolation. The streaming and collision processes are exactly the same as those in the standard LBE model, i.e., the single particle distribution function f_a , as a function of position x and time t , evolves according to the following rule:

$$f_a(x + e_a \delta_t, t + \delta_t) - f_a(x, t) = -\frac{f_a(x, t) - f_a^{eq}(x, t)}{\tau}, \quad (1)$$

where δ_t is the time step and τ is the nondimensional relaxation time. Here the single relaxation time BGK model⁵ has been used for the collision operator. The discrete velocities e_a 's are defined as:

$$e_0 = 0, \quad e_1 = c, \quad e_2 = -c, \quad (2)$$

where the streaming velocity c relates to the sound speed c_s by $c = \sqrt{3}c_s$. The equilibrium distribution function f_a^{eq} has the following form:

$$f_a^{eq} = w_a \rho \left[1 + \frac{3(e_a u)}{c^2} + \frac{9(e_a u)^2}{2c^4} - \frac{3u^2}{2c^2} \right], \quad (3)$$

where $w_0 = 2/3$, and $w_1 = w_2 = 1/6$. The macroscopic density ρ and velocity u are related to the distribution function by:

$$\rho = f_0 + f_1 + f_2, \quad (4)$$

$$\rho u = (f_1 - f_2)c. \quad (5)$$

In a standard LBE scheme, all the grid intervals $\Delta_{i+\frac{1}{2}}$ are equal to the streaming length $\delta_x \equiv c\delta_t$. Consequently, at the next time step, the distribution functions at all grid nodes are automatically determined by the evolution equation (1). This restriction is removed in the ISLBE scheme. A new interpolation is introduced to reconstruct the distribution function at each grid node. **With the distribution function $f_a(x, t + \delta_t)$ known at $x_i + e_a \delta_t$, it is natural to use the Lagrangian interpolation scheme:**

$$f_a(x_i, t + \delta_t) = \sum_{j=0}^n \prod_{k=0, k \neq j}^n \left(\frac{x_i - x_{i+k\gamma_a} - e_a \delta_t}{x_{i+j\gamma_a} - x_{i+k\gamma_a}} \right) f_a(x_{i+j\gamma_a} + e_a \delta_t, t + \delta_t), \quad (6)$$

where n is the order of the interpolation scheme, and the coefficients γ_a 's are chosen to be $\gamma_1 = -1$ and $\gamma_2 = 1$.

The choice of the interpolation points is somewhat arbitrary. In general, they should be chosen as close to x_i as possible to minimize the interpolation error. Also, in order to meet the upwind condition and to maximize the stability of the scheme, it is always desirable to avoid using those points located at downstream x_i . In the following text, it is further assumed that $\delta_x < \Delta_{i+\frac{1}{2}}$ for all i 's. Under this assumption, the set of interpolation points in Eq. (6) is a reasonable choice. Also note that interpolation is not necessary for the distribution function f_0 .

In summary, the ISLBE scheme works as follows:

1. Given initial values of $\rho(x_i, 0)$ and $u(x_i, 0)$;
2. Evolve the collision and streaming steps,

$$f_a(x_i + e_a \delta_t, t + \delta_t) - f_a(x_i, t) = -\frac{f_a(x_i, t) - f_a^{eq}(x_i, t)}{\tau};$$

3. Interpolate

$$f_a(x_i, t + \delta_t) = \sum_{j=0}^n \prod_{k=0, k \neq j}^n \left(\frac{x_i - x_{i+k\gamma_a} - e_a \delta_t}{x_{i+j\gamma_a} - x_{i+k\gamma_a}} \right) f_a(x_{i+j\gamma_a} + e_a \delta_t, t + \delta_t);$$

4. Return to step 2 until a termination condition is satisfied.

3. Error Analysis

To carry out the error analysis, let us first combine the collision, streaming, and interpolation steps and rewrite the evolution equation of the ISLBE scheme as:

$$f_a(x_i, t + \delta_t) = \sum_{j=0}^n \prod_{k=0, k \neq j}^n \left(\frac{x_i - x_{i+k\gamma_a} - e_a \delta_t}{x_{i+j\gamma_a} - x_{i+k\gamma_a}} \right) \times \left[\left(1 - \frac{1}{\tau} \right) f_a(x_{i+j\gamma_a}, t) + \frac{1}{\tau} f_a^{eq}(x_{i+j\gamma_a}, t) \right]. \quad (7)$$

Furthermore, we perform Taylor expansions, at (x_i, t) , of all the distribution functions in Eq. (7) and retain terms up to $O(\delta_t^2)$ and $O(\Delta_{i+\frac{1}{2}}^{n+1})$. The resulting governing equation becomes:

$$f_a + \frac{\partial f_a}{\partial t} \delta_t + \frac{1}{2} \frac{\partial^2 f_a}{\partial t^2} \delta_t^2 = \sum_{m=0}^n (-1)^m \frac{(e_a \delta_t)^m}{m!} \left[\left(1 - \frac{1}{\tau} \right) \frac{\partial^m f_a}{\partial x^m} + \frac{1}{\tau} \frac{\partial^m f_a^{eq}}{\partial x^m} \right] + \sum_{j=0}^n \prod_{k=0, k \neq j}^n \left(\frac{x_i - x_{i+k\gamma_a} - e_a \delta_t}{x_{i+j\gamma_a} - x_{i+k\gamma_a}} \right) \frac{(x_{i+j\gamma_a} - x_i)^{n+1}}{(n+1)!} \times \left[\left(1 - \frac{1}{\tau} \right) \frac{\partial^{n+1} f_a}{\partial x^{n+1}} + \frac{1}{\tau} \frac{\partial^{n+1} f_a^{eq}}{\partial x^{n+1}} \right]. \quad (8)$$

In the following section, the Chapman–Enskog expansion will be applied to Eq. (8) for both linear (1st order) and quadratic (2nd order) interpolation schemes. In the Chapman–Enskog expansion, the distribution function and the time derivative are expanded as:

$$f_a = f_a^{(0)} + \delta_t f_a^{(1)} + \delta_t^2 f_a^{(2)} + O(\delta_t^3) \quad (9)$$

$$\partial_t = \partial_{t_0} + \delta_t \partial_{t_1} + \delta_t^2 \partial_{t_2} + O(\delta_t^3). \quad (10)$$

Here we assume δ_t is of the order of the Knudsen number. It will be shown that the quadratic interpolation is essential to recover the Navier–Stokes equation, while the linear interpolation will inevitably introduce numerical diffusivity and viscosity.

3.1. Quadratic interpolation

For the quadratic interpolation scheme ($n = 2$), Eq. (8) becomes:

$$\begin{aligned} & \frac{\partial f_a}{\partial t} \delta_t + \frac{1}{2} \frac{\partial^2 f_a}{\partial t^2} \delta_t^2 + e_a \delta_t \left[\left(1 - \frac{1}{\tau}\right) \frac{\partial f_a}{\partial x} + \frac{1}{\tau} \frac{\partial f_a^{eq}}{\partial x} \right] \\ & - \frac{1}{2} (e_a \delta_t)^2 \left[\left(1 - \frac{1}{\tau}\right) \frac{\partial^2 f_a}{\partial x^2} + \frac{1}{\tau} \frac{\partial^2 f_a^{eq}}{\partial x^2} \right] \\ & = -\frac{f - f^{eq}}{\tau} + \frac{1}{6} (e_a \delta_t)^3 [(r_{i+\frac{1}{2}\gamma_a} - 1)(r_{i+\frac{\gamma_a}{2}} + r_{i+\frac{3}{2}\gamma_a} - 1) - 1] \\ & \times \left[\left(1 - \frac{1}{\tau}\right) \frac{\partial^3 f_a}{\partial x^3} + \frac{1}{\tau} \frac{\partial^3 f_a^{eq}}{\partial x^3} \right], \end{aligned} \quad (11)$$

where $r_{i+\frac{1}{2}\gamma_a} = \Delta_{i+\frac{1}{2}\gamma_a} / \delta_x$.

The zero order Chapman–Enskog expansion simply leads to:

$$f_a^{(0)} = f_a^{eq}. \quad (12)$$

The first order equation of the Chapman–Enskog expansion is:

$$\left(\frac{\partial}{\partial t_0} + e_a \frac{\partial}{\partial x} \right) f_a^{(0)} = -\frac{f_a^{(1)}}{\tau}. \quad (13)$$

The second order equation of the Chapman–Enskog expansion is:

$$\frac{\partial f_a^{(0)}}{\partial t_1} + \left(\frac{\partial}{\partial t_0} + e_a \frac{\partial}{\partial x} \right) f_a^{(1)} + \frac{1}{2} \frac{\partial^2 f_a^{(0)}}{\partial t_0^2} - \frac{1}{2} (e_a)^2 \frac{\partial^2 f_a^{(0)}}{\partial x^2} - \frac{1}{\tau} e_a \frac{\partial f_a^{(1)}}{\partial x} = -\frac{f_a^{(2)}}{\tau}. \quad (14)$$

Substituting Eq. (13) into the last term on left hand side of Eq. (14), the second order equation of the Chapman–Enskog expansion becomes:

$$\frac{\partial f_a^{(0)}}{\partial t_1} + \left(\frac{\partial}{\partial t_0} + e_a \frac{\partial}{\partial x} \right) f_a^{(1)} + \frac{1}{2} \left(\frac{\partial}{\partial t_0} + e_a \frac{\partial}{\partial x} \right)^2 f_a^{(0)} = -\frac{f_a^{(2)}}{\tau}. \quad (15)$$

Equations (12), (13), and (15) are exactly the same as those in the Chapman–Enskog expansion of the standard lattice-Boltzmann equation.⁸ From here, it is straightforward to prove that this approximation recovers the Navier–Stokes equation as long as the equilibrium distribution function is given by Eq. (3).

The only difference between the Chapman–Enskog expansion of the ISLBE and the LBE model is the last term on the right hand side of Eq. (11). In the above derivation, it was assumed that this term is of the order of δ_t^3 . This is always true for an LBE scheme, because $r_{i+\frac{1}{2}\gamma_a} = 1$ for all i 's. The leading error of this term simply reduces to:

$$-\frac{1}{6}(e_a\delta_t)^3\frac{\partial^3 f_a^{eq}}{\partial x^3}. \quad (16)$$

For an ISLBE scheme, however, the magnitude of this term depends on $r_{i+\frac{1}{2}\gamma_a}$, the ratio of the grid size to the free streaming length. To ensure the recovery of the Navier–Stokes equation, the flow domain must be discretized appropriately to keep this term of the order of δ_t^3 . In other words, the grid size should be chosen to keep the magnitude of

$$\frac{1}{6}[(r_{i+\frac{1}{2}\gamma_a} - 1)(r_{i+\frac{1}{2}\gamma_a} + r_{i+\frac{3}{2}\gamma_a} - 1) - 1]\frac{\partial^3 f_a^{eq}}{\partial x^3} \quad (17)$$

of the order of one. Otherwise, numerical errors become significant.

In regions where the flow experiences a large spatial variation or $\partial^3 f_a^{eq}/\partial x^3$ is not a small quantity, the grid size should be kept at the same order of the streaming length δ_x . This restriction has been observed in our previous numerical simulations of flow in a sudden expansion.¹³ When an r of 8 is used in the recirculation zone, the global interpolation error in velocity is 1.68 percent, and the difference in the streamlines is noticeable.

On the other hand, in regions where flow experiences a small spatial variation or $\partial^3 f_a^{eq}/\partial x^3 \ll 1$, the restriction on the grid size can be relaxed considerably. For example, when a non-uniform mesh is used in simulation of flow in a sudden expansion, r can be increased to 32 downstream, and the result is still in good agreement with the standard LBE simulation.

3.2. Linear interpolation

It has been shown above that the ISLBE scheme recovers the Navier–Stokes equation as long as a quadratic interpolation scheme is used and the flow domain is discretized appropriately. It will be interesting to check the error that a linear interpolation will bring into the ISLBE scheme, since in practice a linear interpolation is more efficient due to its simplicity. Unfortunately, as shown below, a linear interpolation will inevitably introduce a significant numerical error in approximating the Navier–Stokes equation.

For a linear interpolation scheme ($n = 1$), Eq. (8) becomes:

$$\begin{aligned} & \frac{\partial f_a}{\partial t} \delta_t + \frac{1}{2} \frac{\partial^2 f_a}{\partial t^2} \delta_t^2 + e_a \delta_t \left[\left(1 - \frac{1}{\tau}\right) \frac{\partial f_a}{\partial x} + \frac{1}{\tau} \frac{\partial f_a^{eq}}{\partial x} \right] \\ &= -\frac{f - f^{eq}}{\tau} + \frac{1}{2} (e_a \delta_t)^2 r_{i+\frac{1}{2}\gamma_a} \left[\left(1 - \frac{1}{\tau}\right) \frac{\partial^2 f_a}{\partial x^2} + \frac{1}{\tau} \frac{\partial^2 f_a^{eq}}{\partial x^2} \right]. \end{aligned} \quad (18)$$

The zeroth and first order Chapman–Enskog expansions of Eq. (18) have the same form as Eqs. (12) and (13), respectively. Therefore, the ISLBE scheme can still recover to the Euler equation.

The second order Chapman–Enskog expansion of Eq. (18), however, now has the following form:

$$\begin{aligned} & \frac{\partial f_a^{(0)}}{\partial t_1} + \left(\frac{\partial}{\partial t_0} + e_a \frac{\partial}{\partial x} \right) f_a^{(1)} + \frac{1}{2} \left(\frac{\partial}{\partial t_0} + e_a \frac{\partial}{\partial x} \right)^2 f_a^{(0)} \\ &= -\frac{f_a^{(2)}}{\tau} + \frac{1}{2} (r_{i+\frac{1}{2}\gamma_a} - 1) e_a^2 \frac{\partial^2 f_a^{(0)}}{\partial x^2}. \end{aligned} \quad (19)$$

In the Chapman–Enskog expansion of the standard LBE models, the last term at the right hand side of Eq. (19) is equal to zero because $r_{i+\frac{1}{2}\gamma_a} = 1$. For the ISLBE scheme, this term is no longer zero and it inevitably leads to numerical diffusivity and viscosity.

Assuming $r_{i+\frac{1}{2}\gamma_a} \approx \bar{r}_i$, summation of Eq. (19) over all a 's leads to

$$\frac{\partial \rho}{\partial t_1} = \frac{1}{2} (\bar{r}_i - 1) \frac{\partial^2}{\partial x^2} [(c_s^2 + u^2) \rho] \approx \frac{1}{2} (\bar{r}_i - 1) c_s^2 \frac{\partial^2 \rho}{\partial x^2}. \quad (20)$$

The right hand side of Eq. (20) represents a spurious diffusion term because the corresponding diffusivity depends on grid size.

Multiplying Eq. (19) by e_a and summing over a , we have:

$$\frac{\partial \rho u}{\partial t_1} = \frac{\partial}{\partial x} \left(\rho \nu \frac{\partial u}{\partial x} \right) + \frac{1}{2} (\bar{r}_i - 1) \frac{\partial^2}{\partial x^2} (\rho u), \quad (21)$$

where

$$\nu = \frac{2\tau - 1}{6} \frac{\delta_x^2}{\delta_t}$$

is the fluid viscosity. The last term of Eq. (21) represents a spurious viscous term because the corresponding viscosity depends on grid size.

The inevitable numerical error due to the first order interpolation scheme has also been observed in numerical simulations of flow in a sudden expansion. The global error is quite large (3.17 percent) even for the grid size chosen to be twice the streaming length. Obviously, a linear interpolation scheme is not appropriate for the ISLBE scheme.

4. Conclusion

The numerical error in the interpolation-supplemented lattice-Boltzmann BGK scheme has been theoretically analyzed. The numerical error is found to depend on the interpolation scheme and the way the flow domain is discretized. A linear interpolation scheme is in general not appropriate for the ISLBE scheme, since it inevitably introduces excessive numerical diffusion and viscosity. A quadratic interpolation scheme warrants the recovery of the Navier–Stokes equation, as long as the flow domain is appropriately discretized. The analysis is consistent with previous numerical experiments.

It is worthwhile at this point to compare the ISLBE scheme with other LBE-based schemes intending to extend the LBE method to variable meshes. Two types of efforts can be found along this line: the finite volume approach (FVLBE) and the finite difference approach (FDLBE). The FVLBE scheme^{16,17} results from the marriage of the LBE method with the finite volume technique. In the FVLBE scheme, a course-grained distribution function in each macrocell is updated according to (1) the net flux of the distribution function across the cell boundaries, and (2) the relaxation of the distribution function to the equilibrium state in each cell. The FVLBE is an analogue of the gas-kinetic finite volume method.¹⁸ The difference is that the FVLBE solves the Boltzmann equation only for a finite set of velocities, while the latter solves the Boltzmann equation for all microscopic velocities. Like the ISLBE, the FVLBE also requires an interpolation for the reconstruction step and its accuracy strongly depends on the interpolation scheme as well. Because the reconstruction is necessary along the entire cell boundaries, the FVLBE requires more computation than the ISLBE for multi-dimensional problems.

The FDLBE scheme¹² is formulated in the framework of the finite difference method, in which the Boltzmann equation is solved by standard finite difference schemes. This approach originates from the observation that the LBE can be regarded as a first order upwind finite-difference discretization of the Boltzmann equation.^{19,20} The LBE is of second order accuracy only because the second order truncating error behaves exactly as the physical viscosity and can be absorbed into the physical viscosity when a uniform grid is used. Because this absorption is no longer valid for a variable grid, the FDLBE uses a second order central-difference scheme to eliminate the second order truncating error. Unfortunately, the introduction of the central-difference scheme sacrifices the upwind condition and reduces the stability of the LBE scheme.

Finally, it should be pointed out that the analysis in this work is only appropriate for the LBE scheme. The third order interpolation is not necessary for directly solving the Navier–Stokes equation.

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